Dynamical Systems Model of Identity Braids: Energy Landscapes and Tension Dynamics

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Abstract

We develop a dynamical systems model of identity formation based on the interaction of four identity components: self-perception, aspirational identity, social perception, and reality constraints. The interaction of these variables produces tension structures that can be represented as an energy landscape.

Identity trajectories evolve through gradient dynamics within this landscape, while discrete selection events introduce path dependence and topological braid crossings.

The framework provides a mathematical basis for understanding emotional tension, identity stability, and identity transitions.

1 Introduction

Identity formation is traditionally described using narrative, psychological, or sociological frameworks. These approaches emphasize life stories, developmental stages, and social roles as the primary drivers of identity formation. While such perspectives capture many important aspects of identity, they often lack a formal description of how identity states evolve over time.

In this paper we propose a complementary dynamical systems framework in which identity is modeled as a trajectory through a tension landscape defined by interacting identity components. Rather than treating identity as a static label or narrative description, we represent it as an evolving configuration of variables that continuously interact with one another and with external context.

The model builds on the conceptual framework introduced in *Identity Tension and Emotional Structure Recognition* (Paper 1), where emotional states were interpreted as tensions between identity components. In that work we argued that emotions emerge when identity variables diverge from one another, producing subjective experiences of psychological strain.

The present paper develops a mathematical representation of this idea. We show that interactions among identity components can be described using an energy-based dynamical system in which identity trajectories move through a tension landscape. Within this landscape, stable identity configurations correspond to low-energy attractors, while identity conflict and emotional tension correspond to higher-energy states.

This perspective also naturally introduces path dependence in identity development. Certain events can produce discrete transitions in the identity trajectory, permanently altering future dynamics. When viewed over time, these trajectories form braided structures that capture the historical accumulation of identity decisions and constraints.

Together, these elements provide a unified dynamical framework for modeling identity formation, emotional tension, and identity transitions..

2 Identity State Space

We represent identity as a state vector composed of four interacting components:

$$X(t) = (S(t), D(t), P(t), R(t))$$

where each variable represents a distinct dimension of identity experience.

- $S(t)$ represents **self-perception**, the individual's internal model of who they currently are.
- $D(t)$ represents **aspirational identity**, the identity state the individual desires or believes they ought to become.
- $P(t)$ represents **social perception**, the identity that others attribute to the individual through social interaction and cultural interpretation.
- $R(t)$ represents **reality constraints**, which include material conditions, abilities, and environmental limitations that restrict possible identity states.

These four variables capture the fundamental tensions that shape human identity. Self-perception reflects the internal narrative of identity, aspirational identity represents directional motivation, social perception introduces external feedback, and reality constraints represent the practical boundaries within which identity must operate.

Identity formation can therefore be understood as the evolution of the state vector $X(t)$ through time as these components interact with one another.

3 Identity Tension Coordinates

Define pairwise tensions

$$\delta_{SD} = S - D$$

$$\delta_{SP} = S - P$$

$$\delta_{SR} = S - R$$

These tensions define coordinates in emotional space.

4 Identity Tension Coordinates

Identity tensions arise when components of the identity state vector diverge from one another. These divergences create psychological strain that individuals experience subjectively as emotional states.

We define three primary tension coordinates that capture the most common sources of identity conflict:

$$\delta_{SD} = S - D$$

$$\delta_{SP} = S - P$$

$$\delta_{SR} = S - R$$

Each of these coordinates represents a distinct form of identity tension.

- **Aspirational tension** (δ_{SD}) measures the distance between the individual's current self-perception and their desired identity. Large values correspond to experiences of ambition, self-dissatisfaction, or motivation for change.
- **Social tension** (δ_{SP}) measures the mismatch between how individuals see themselves and how they are perceived by others. High values often correspond to experiences such as shame, social anxiety, or identity misrecognition.
- **Reality tension** (δ_{SR}) captures the gap between self-perception and external constraints. This form of tension often produces experiences associated with cognitive dissonance or frustration when aspirations conflict with practical limitations.

Together these three tensions define coordinates within a reduced emotional state space. Emotional experiences can therefore be interpreted as configurations of identity tension within this space.

For example:

- High δ_{SD} with low social tension may correspond to ambition or aspiration.
- High δ_{SP} may correspond to shame or social exposure.
- High δ_{SR} may correspond to frustration or cognitive dissonance.

In this framework emotions are not treated as isolated psychological categories, but as regions within a continuous tension space defined by interactions among identity components. .

5 Identity Dynamics

Identity evolves according to gradient dynamics

$$\frac{dX}{dt} = -\nabla E(X)$$

which drives the system toward lower tension configurations.

6 Selection Events

Certain events create discrete transitions in identity:

$$X(t^+) = \Phi(X(t^-), E)$$

These transitions introduce path dependence and create persistent braid structures in identity trajectories.

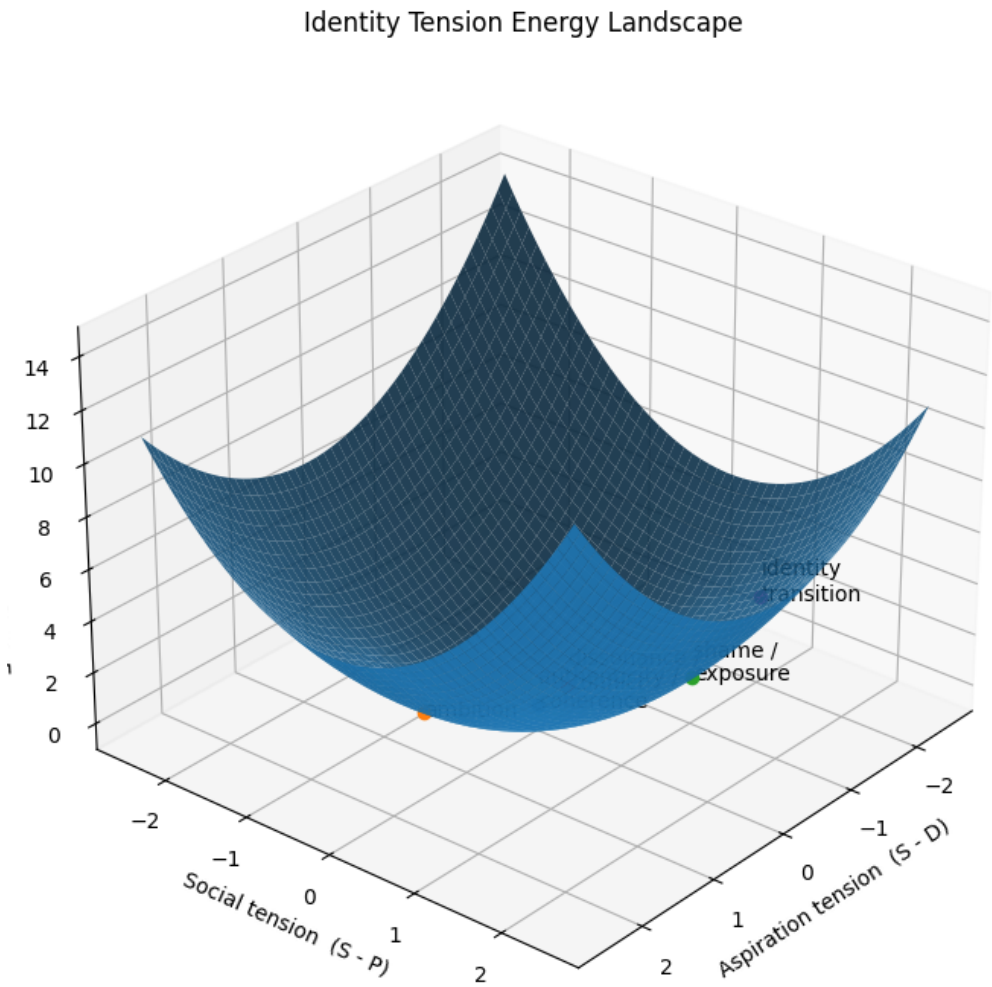

Figure 1: Conceptual identity tension energy landscape in reduced coordinates. Low-energy regions correspond to coherence and authenticity, while increasing aspiration and social tension move the system into regions associated with ambition, shame/exposure, conflict, and identity transition.

7 Stability and Identity Attractors

Stable identities correspond to minima of the identity energy landscape satisfying

$$\nabla E(X) = 0$$

8 Discussion

This dynamical framework provides a mathematical foundation for the identity braid model introduced in Paper 1 and connects emotional structure to energy-based dynamical systems.

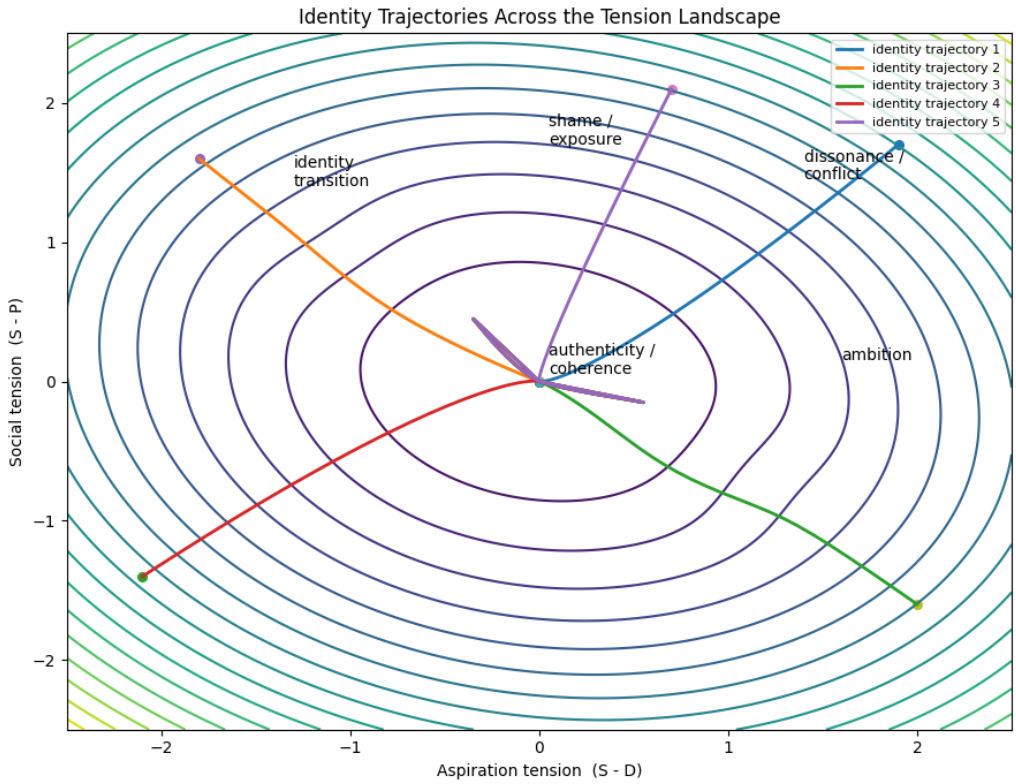

Figure 2: Conceptual trajectories of identity states through the reduced tension landscape. Different initial conditions and discrete selection events produce distinct paths through regions associated with ambition, shame/exposure, conflict, and identity transition, while trajectories tend toward lower-energy coherence basins.

9 Conclusion

Identity formation can be understood as a dynamical system evolving within a tension landscape defined by interacting identity components. This perspective offers a unified framework for understanding emotional dynamics, identity stability, and identity transitions.